Detection Engineering Lab Evidence

Atomic Red Team attack simulation → Sysmon / Windows Security logs → Splunk detection

Item	Detail
Date of testing	Saturday, 31 October 2026
Target host	DESKTOP-STQR2I1 (Windows 11 Home, build 26200, VirtualBox VM, workgroup)
Attack simulation	Atomic Red Team via Invoke-AtomicTest (PowerShell, run as Administrator)
Telemetry	Sysmon (Microsoft-Windows-Sysmon/Operational) and Windows Security log, shipped by Splunk Universal Forwarder
SIEM	Splunk Enterprise, index=main

Summary of results

MITRE ATT&CK	Technique	Atomic test run	Detected in Splunk by
T1082	System Information Discovery	T1082-1	Sysmon EID 1 (systeminfo.exe, parent cmd.exe) and Security EID 4688
T1053.005	Scheduled Task	T1053.005-1 (OnLogon and OnStartup tasks)	Sysmon EID 1 (schtasks.exe /create, parent cmd.exe)
T1547.001	Registry Run Keys / Startup Folder	T1547.001-1 (Reg Key Run)	Sysmon EID 13 (registry value set under ...\CurrentVersion\Run, by reg.exe)
T1136.001	Create Account: Local Account	T1136.001-4 (cmd) and -8 (admin user)	Security EIDs 4720, 4722, 4732, 4726

Each section below follows the same order: (1) the attack executed on the host, then (2) the matching telemetry found in Splunk. All tests were followed by the Atomic cleanup step, and the cleanup is also visible in the evidence.

1. T1082: System Information Discovery

Step 1: Attack executed. `Invoke-AtomicTest T1082 -TestNumbers 1` ran systeminfo and a registry query for the disk enumeration key. The command exited with code 0.

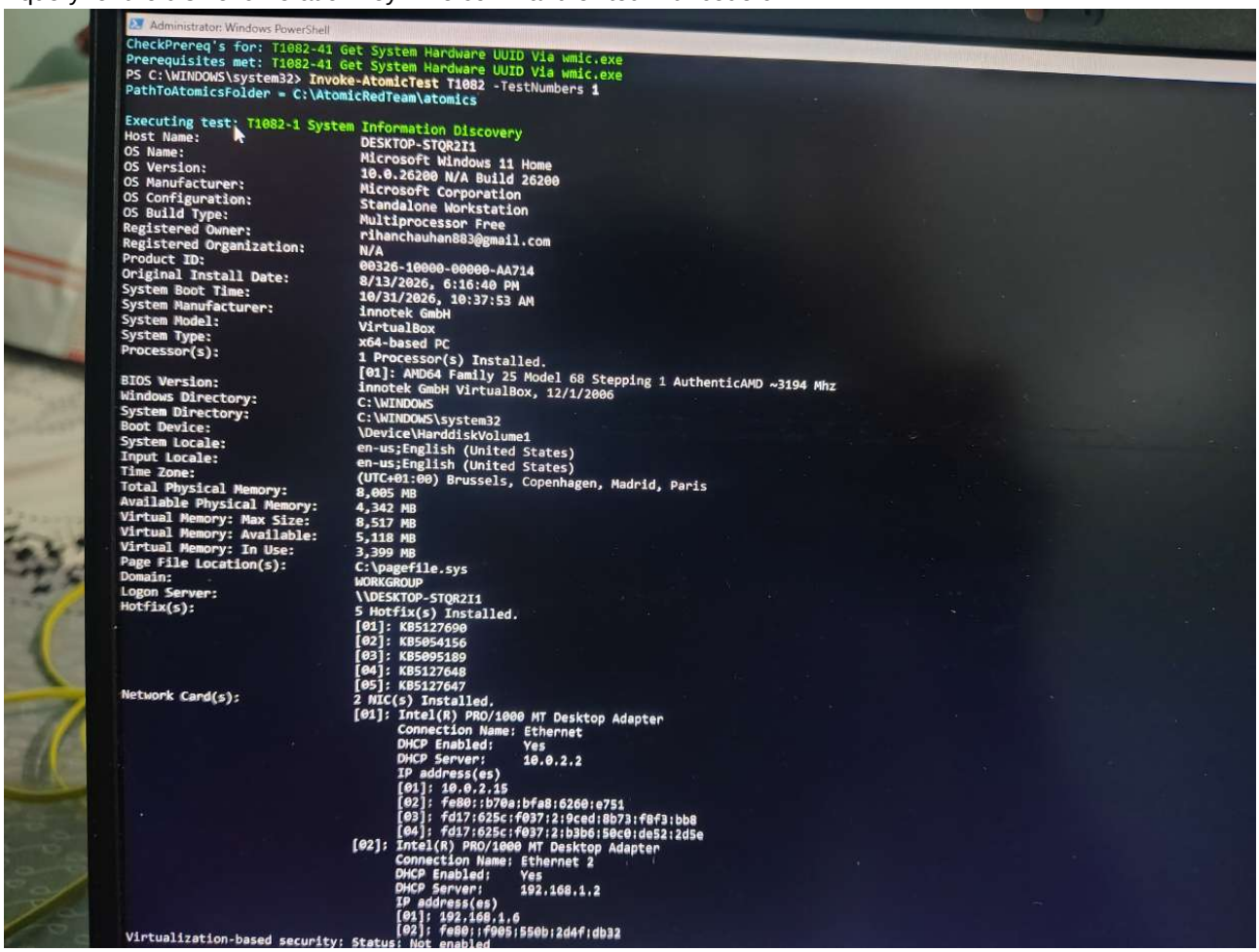


Figure 1.1: Atomic test T1082-1 executing; systeminfo output (host, OS, NICs).

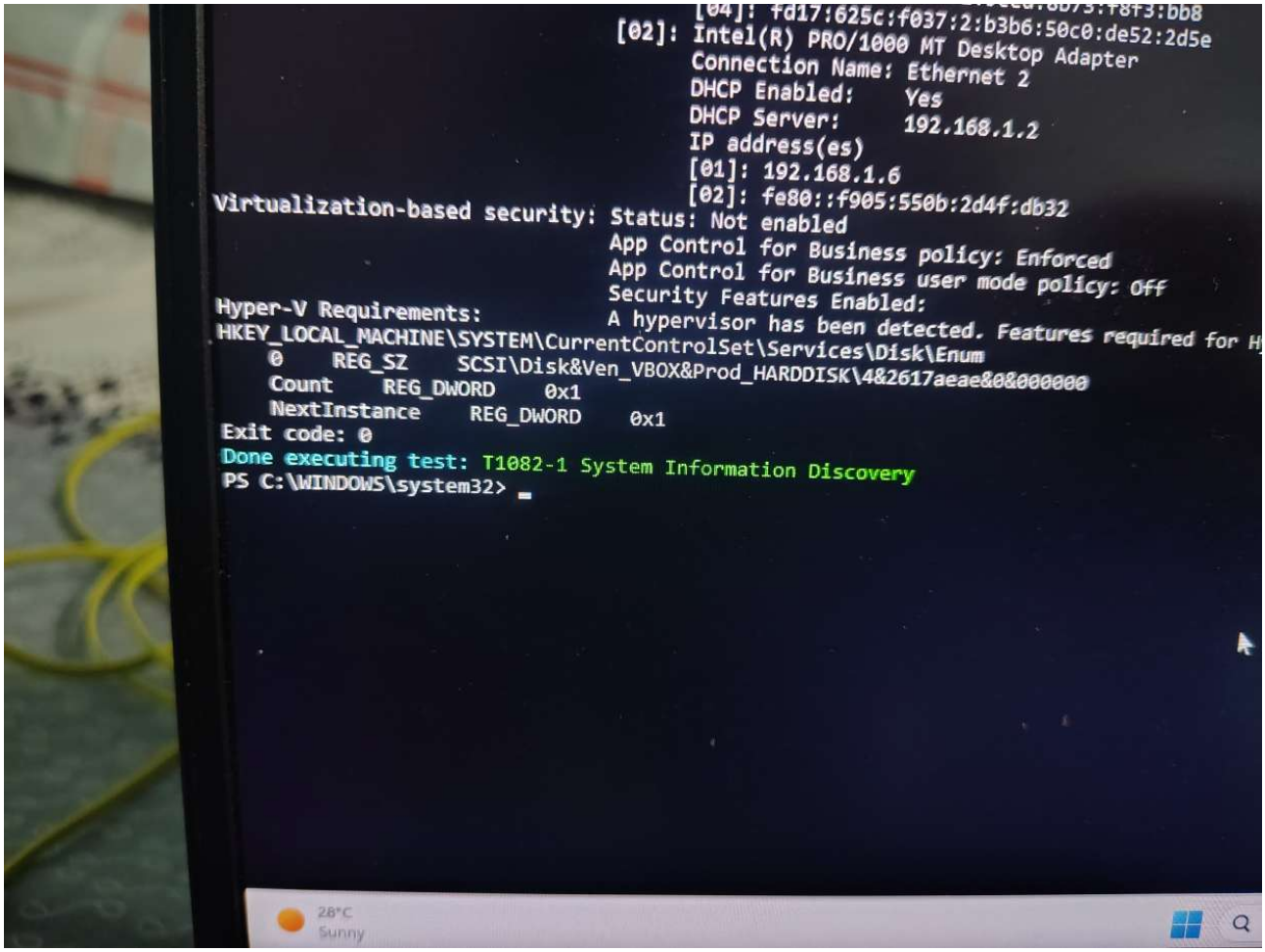


Figure 1.2: End of output (registry query on Services\Disk\Enum), Exit code 0, test completed.

Step 2: Detection in Splunk. Searching for `systeminfo.exe` returned 3 events, from both the Windows Security log and Sysmon. The Sysmon event shows the parent command line `cmd.exe /c systeminfo & reg query HKLM\SYSTEM\CurrentControlSet\Services\Disk\Enum`, run by user rihan.

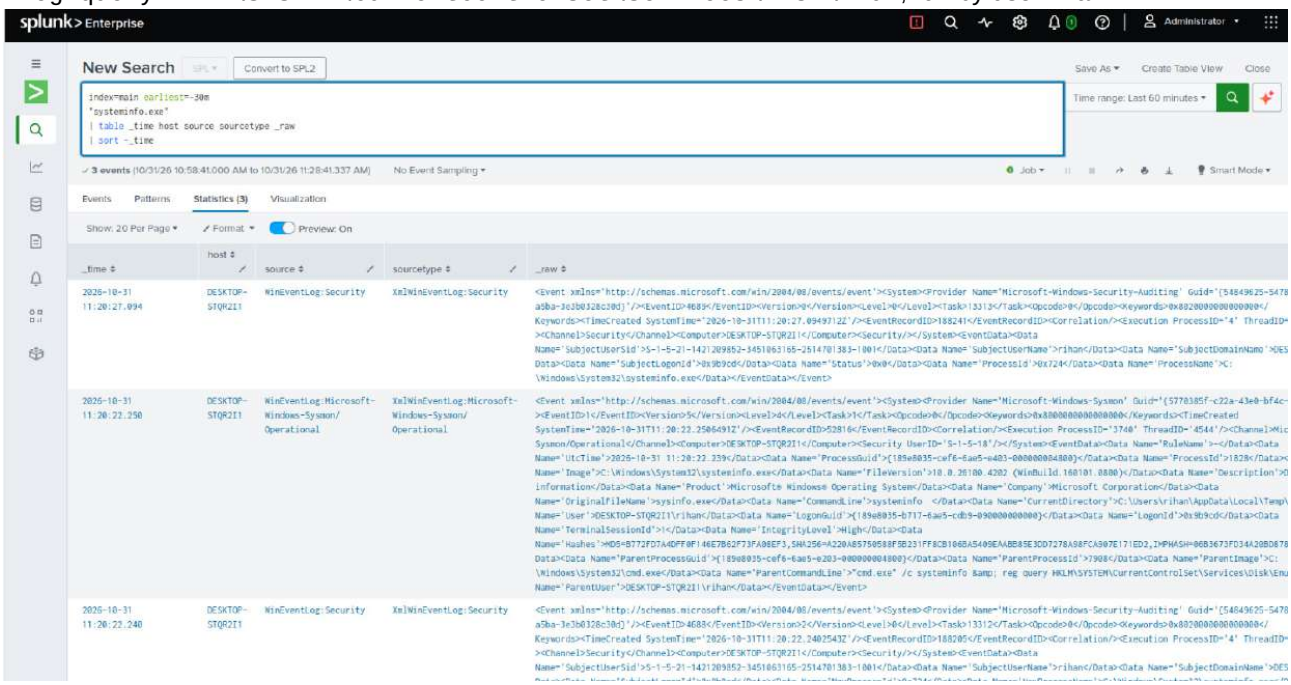


Figure 1.3: Splunk search for systeminfo.exe: Security log (4688) and Sysmon process-creation events.

A broader Sysmon process-creation search (EventCode=1, excluding Splunk's own processes) lists the surrounding activity: cmd.exe, systeminfo.exe, whoami.exe and HOSTNAME.EXE, in time order.

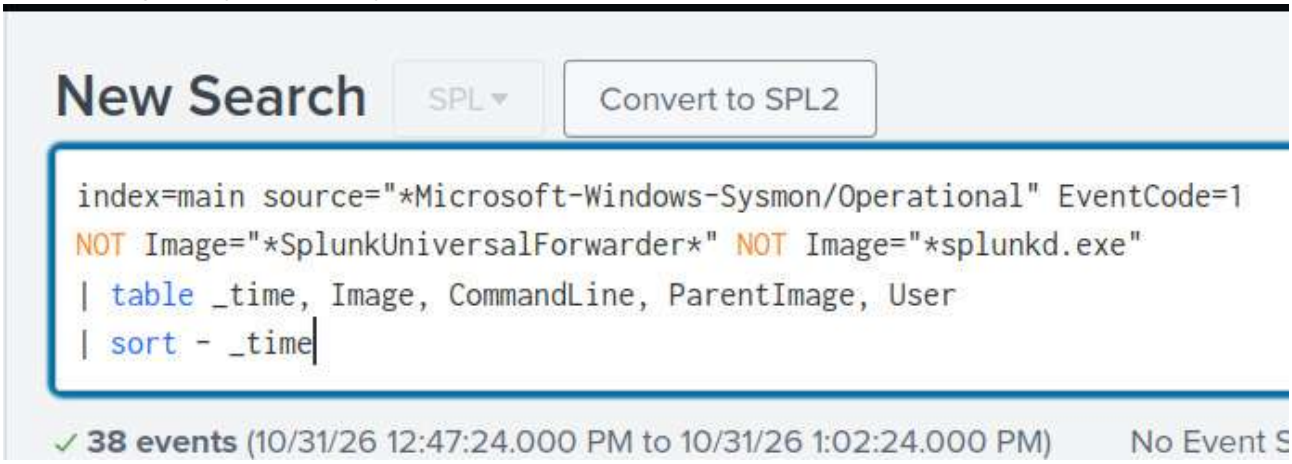


Figure 1.4: SPL used to list Sysmon process creation events, excluding Splunk Universal Forwarder / splunkd.exe noise.

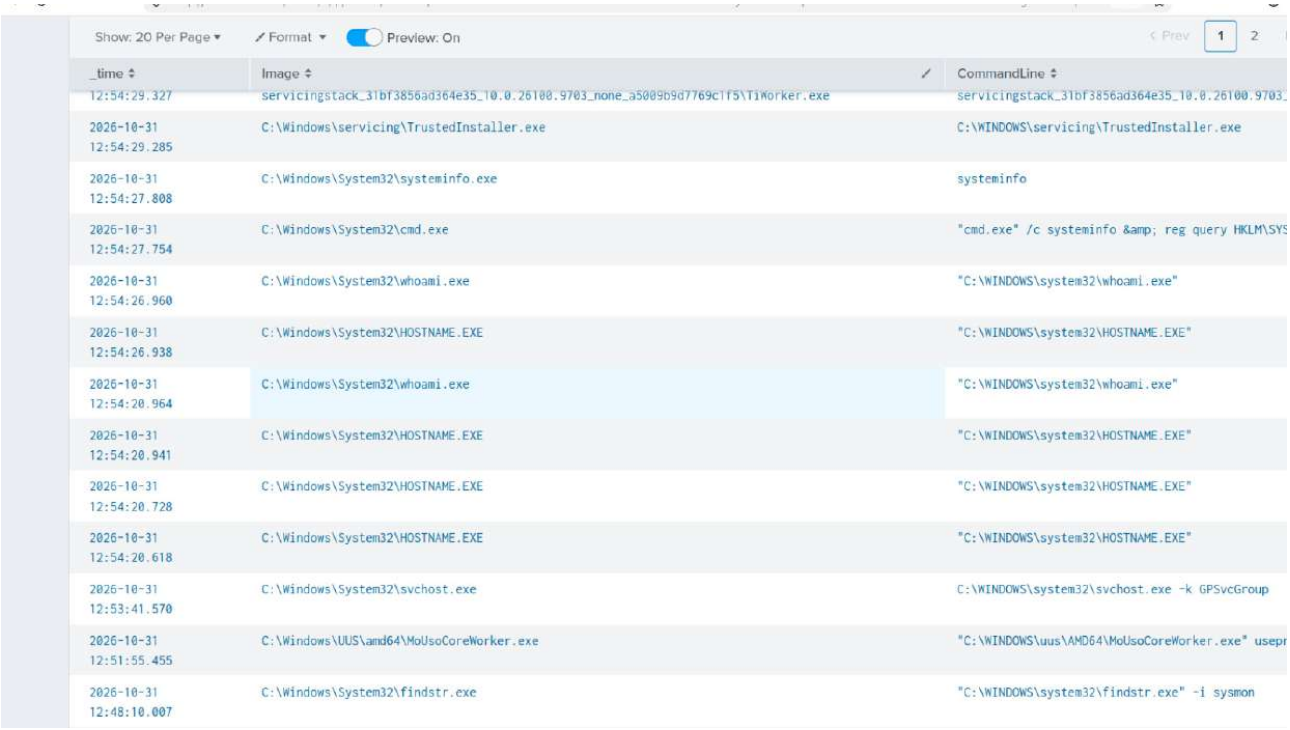


Figure 1.5: Results (38 events): cmd.exe /c systeminfo, systeminfo.exe, whoami.exe, HOSTNAME.EXE.

2. T1053.005: Scheduled Task

Step 1: Attack executed. `Invoke-AtomicTest T1053.005 -TestNumbers 1` created two scheduled tasks, `T1053_005_OnLogon` and `T1053_005_OnStartup`. Both reported SUCCESS with exit code 0. I then confirmed the task exists with `schtasks /query /tn "T1053_005_OnLogon"` (status: Ready).

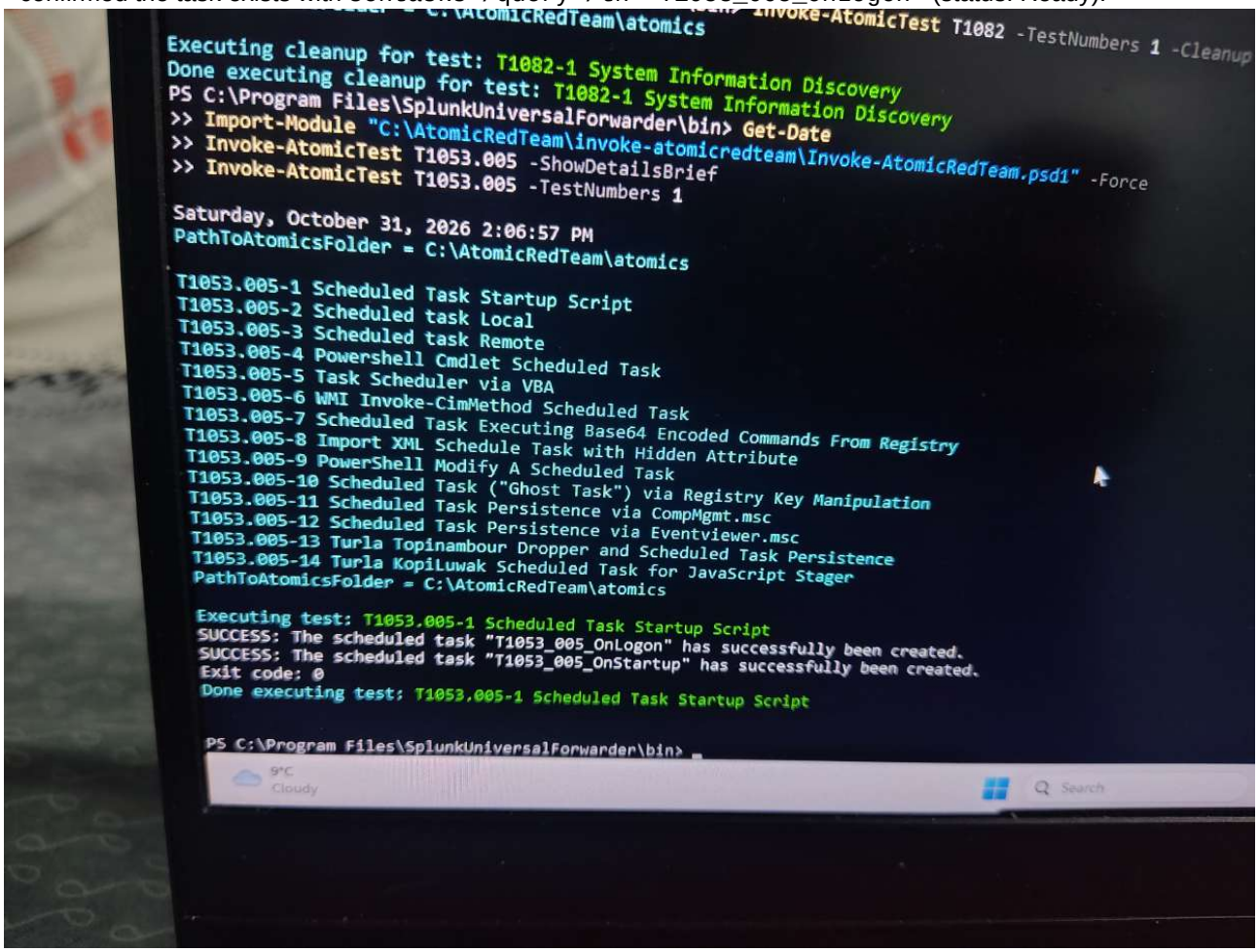


Figure 2.1: Atomic T1053.005-1 creating both scheduled tasks (SUCCESS, Exit code 0).

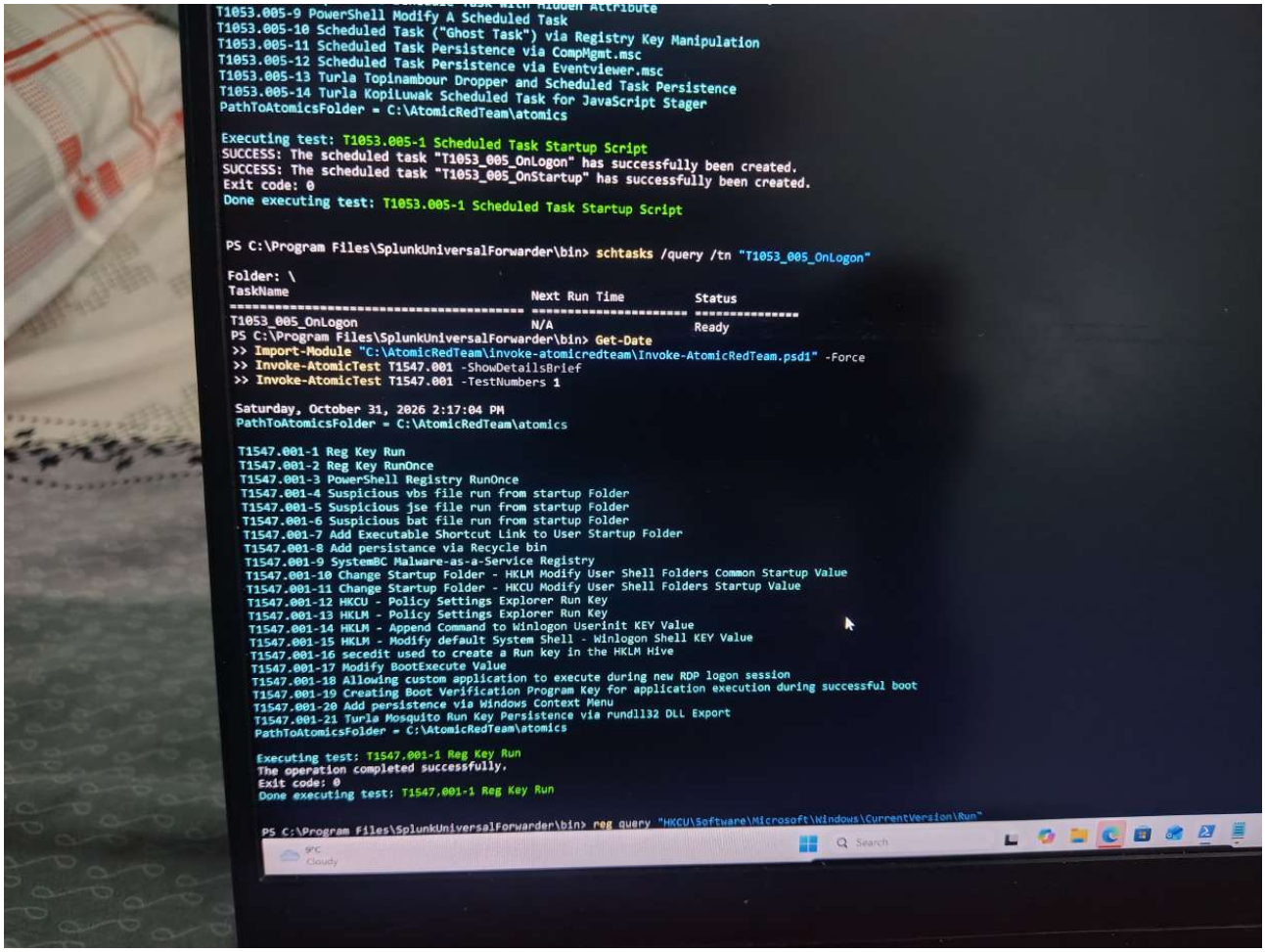


Figure 2.2: schtasks /query confirming T1053_005_OnLogon is present (top half of the screen).

Step 2: Detection in Splunk. Filtering Sysmon process creation on `schtasks.exe` returned 3 events: two `schtasks /create` events (OnLogon with /sc onlogon, OnStartup with /sc onstart /ru system) whose task action is `cmd.exe /c calc.exe`, plus the later /query. Parent process is cmd.exe (create) and powershell.exe (query), user DESKTOP-STQR2I1\rihan.

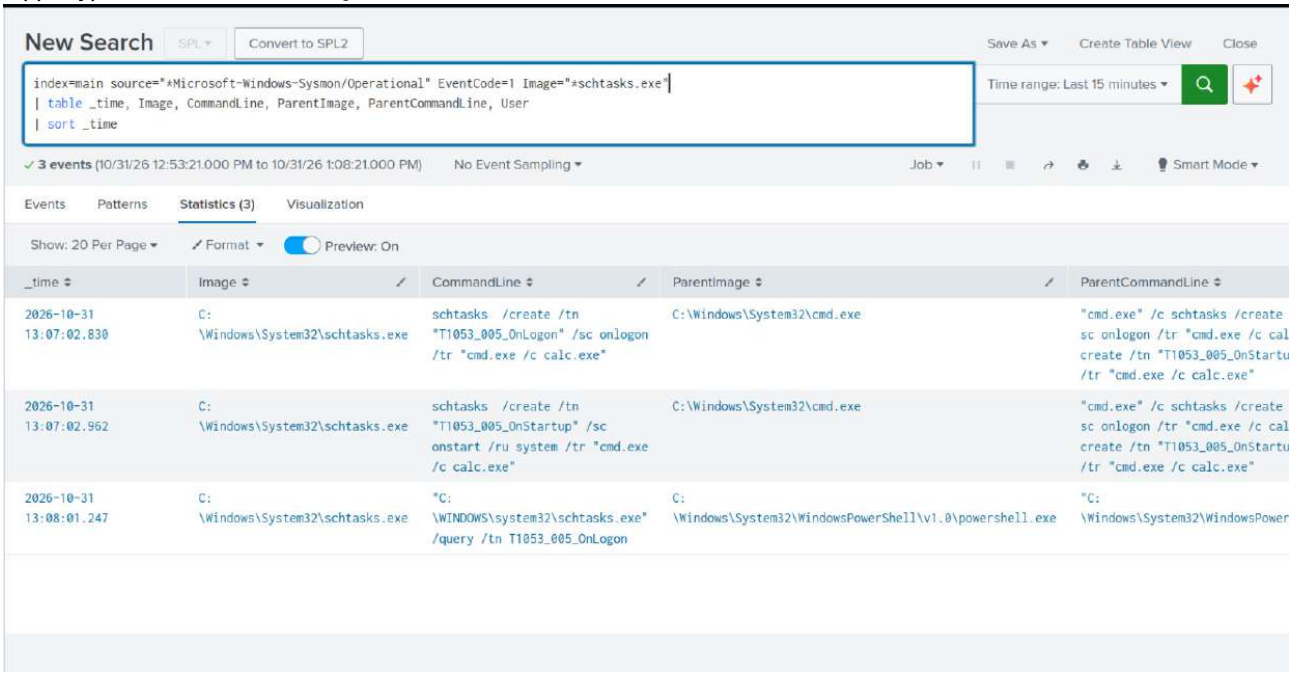


Figure 2.3: Sysmon EID 1 for schtasks.exe: two /create events at 13:07:02 and the /query at 13:08:01.

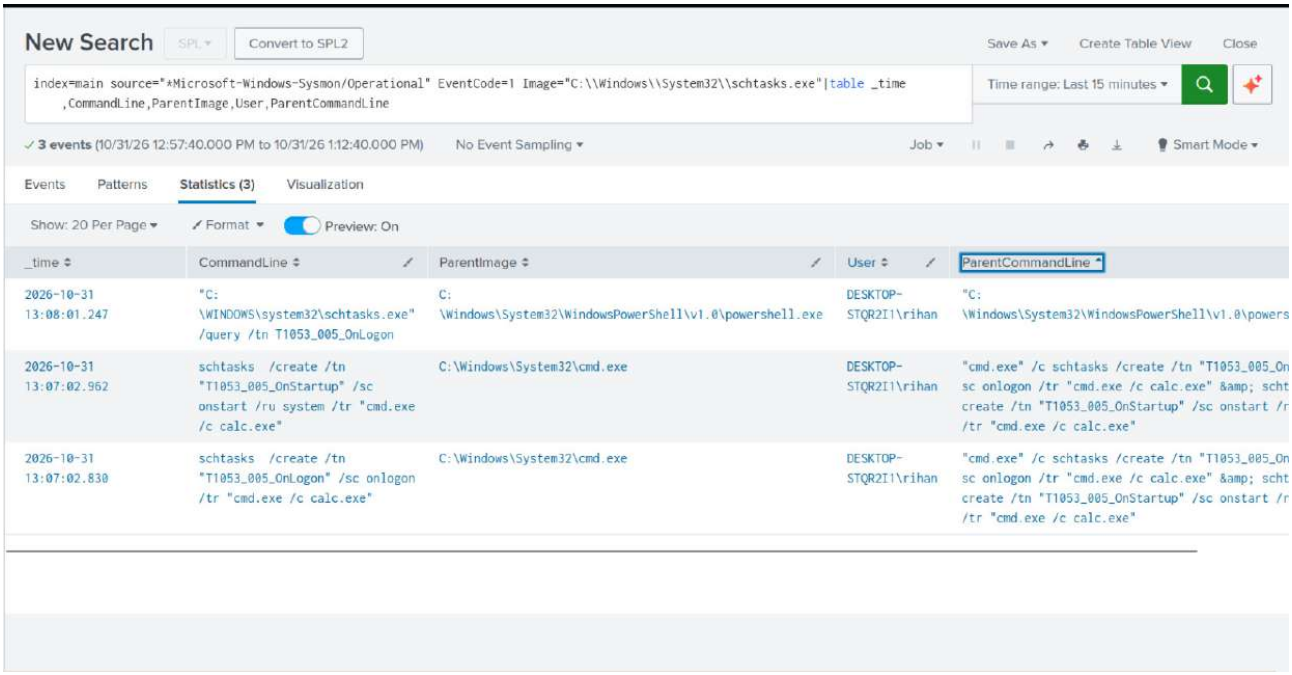


Figure 2.4: Refined search (exact image path) adding the User field and the full ParentCommandLine.

3. T1547.001: Registry Run Keys / Startup Folder

Step 1: Attack executed. `Invoke-AtomicTest T1547.001 -TestNumbers 1` ("Reg Key Run") completed successfully. A follow-up `reg query` of `HKCU\Software\Microsoft\Windows\CurrentVersion\Run` shows the new persistence value **Atomic Red Team** pointing to `C:\Path\AtomicRedTeam.exe`, next to the normal OneDrive and Edge autolaunch entries. The Atomic cleanup was run afterwards.

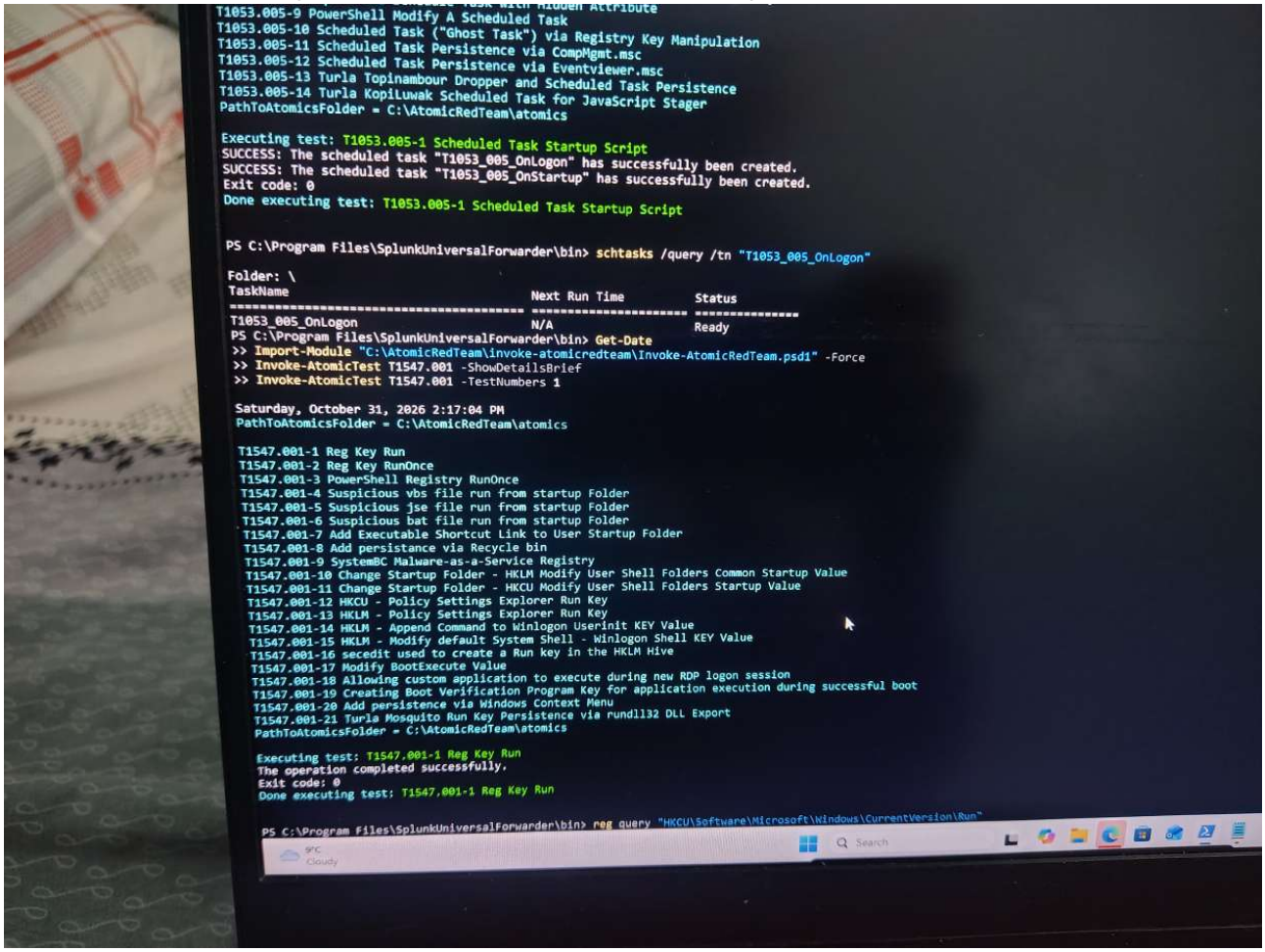


Figure 3.1: T1547.001-1 executing, "The operation completed successfully", Exit code 0 (lower half of the screen).

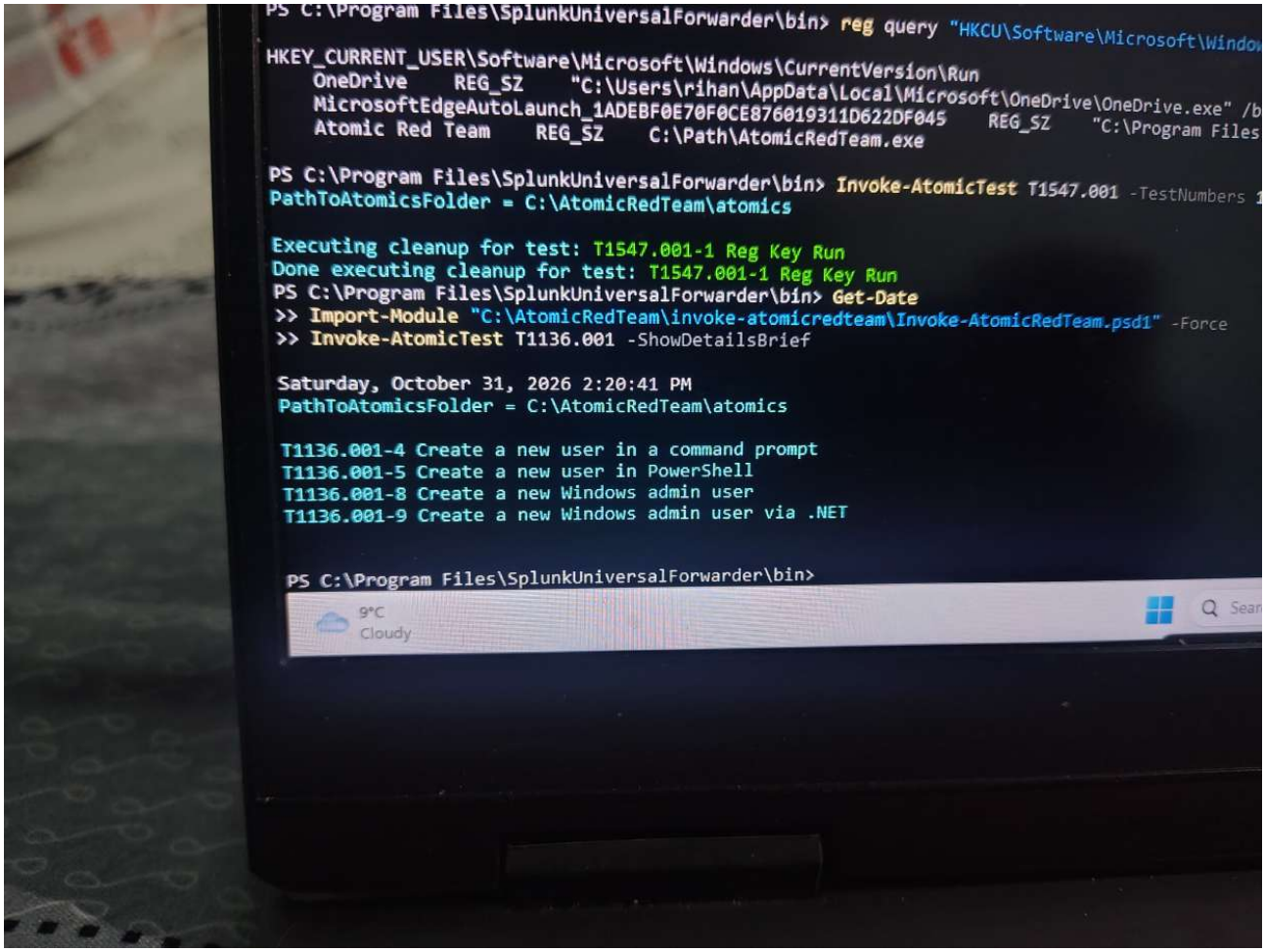


Figure 3.2: reg query of the HKCU Run key showing the "Atomic Red Team" value, followed by the cleanup run.

Step 2: Detection in Splunk. A Sysmon EventCode=13 (registry value set) search on `*\Run*` over 24 hours returned 6 events. Five are Microsoft Edge (msedge.exe) writing its own autolaunch value, which is expected baseline activity. The last, at 13:17:11, is the test: `reg.exe` writing the "Atomic Red Team" value.

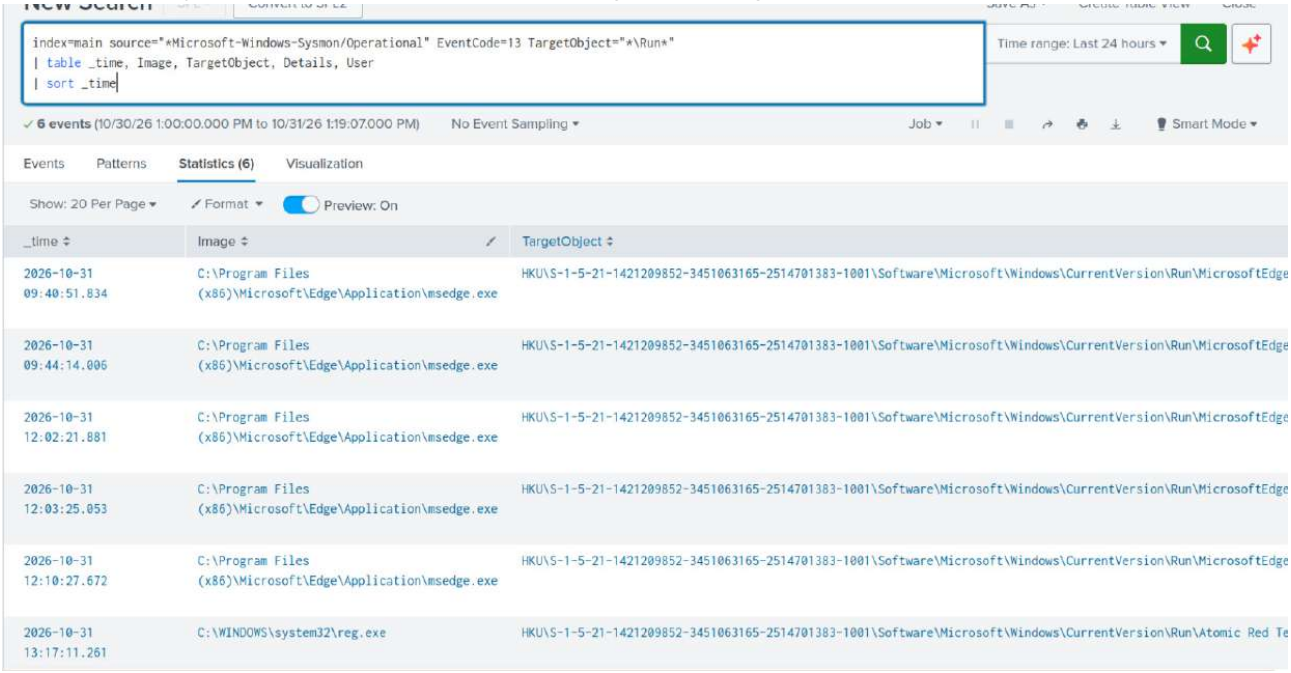


Figure 3.3: Sysmon EID 13 on Run keys: benign msedge.exe entries versus the reg.exe write at 13:17:11.

Drilling into that single event confirms the details: EventType SetValue, Image `C:\WINDOWS\system32\reg.exe`, Details `C:\Path\AtomicRedTeam.exe`, User DESKTOP-STQR2I1\rihan, and Sysmon's rule name T1060,RunKey.

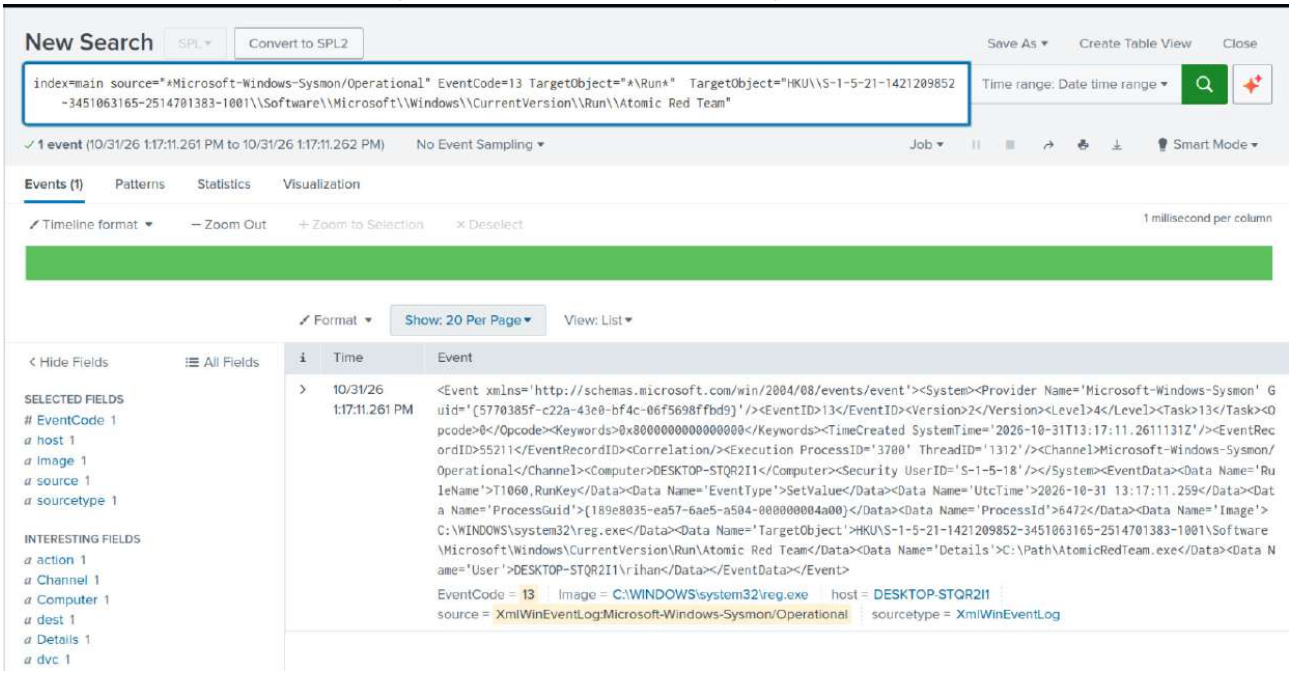


Figure 3.4: Full raw Sysmon event for the Run key write.

4. T1136.001: Create Account: Local Account

Step 1: Attack executed. The available tests were listed with `Invoke-AtomicTest T1136.001 -ShowDetailsBrief`. Test 4 (create a user in a command prompt) created **T1136.001_CMD**, visible in `net user`, and was then removed with `-Cleanup` (the account is gone from the second listing). Test 8 (create a Windows admin user) created **T1136.001_Admin**, which appears in the local Administrators group and was also removed by cleanup.

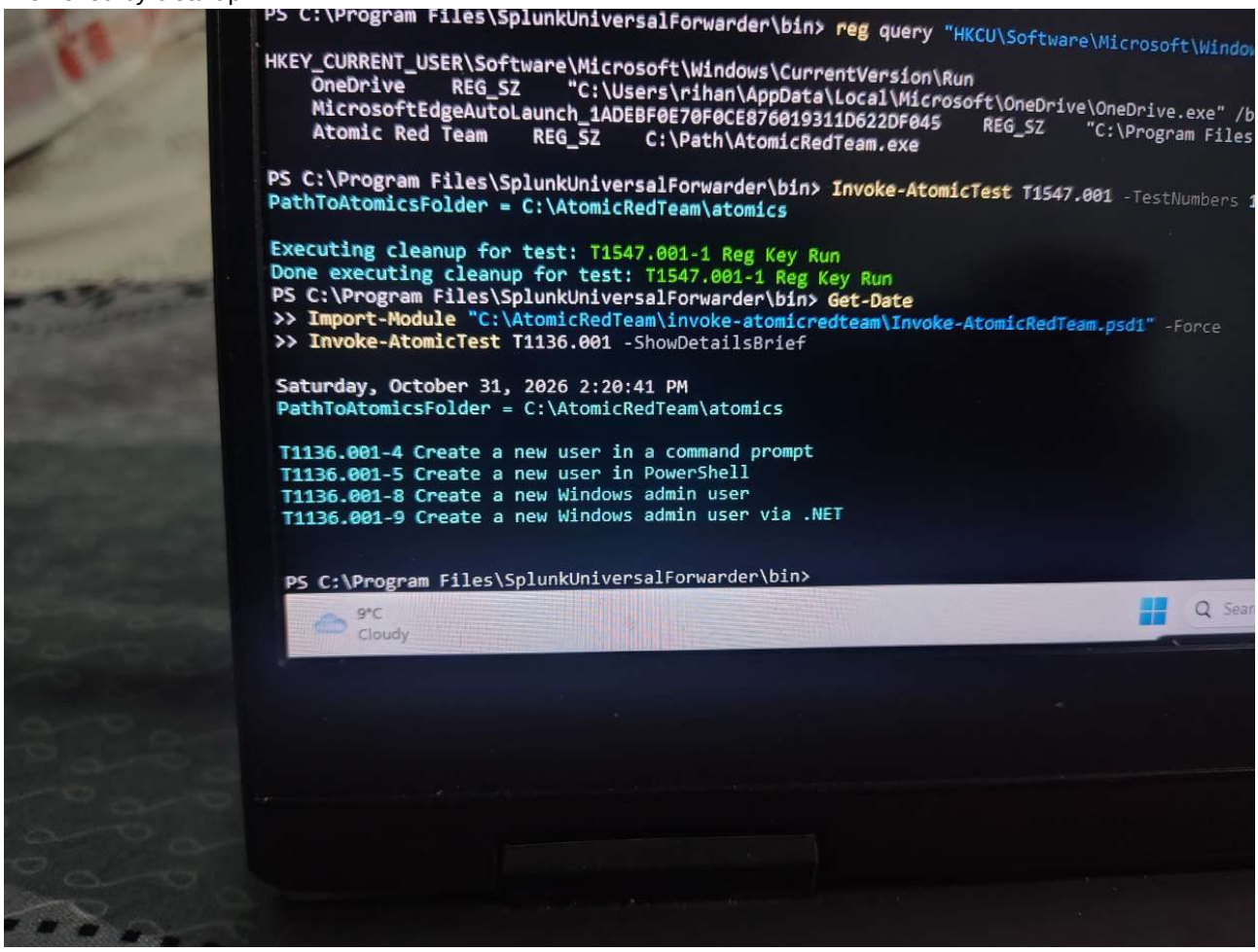


Figure 4.1: Test list for T1136.001 (bottom half of the screen).

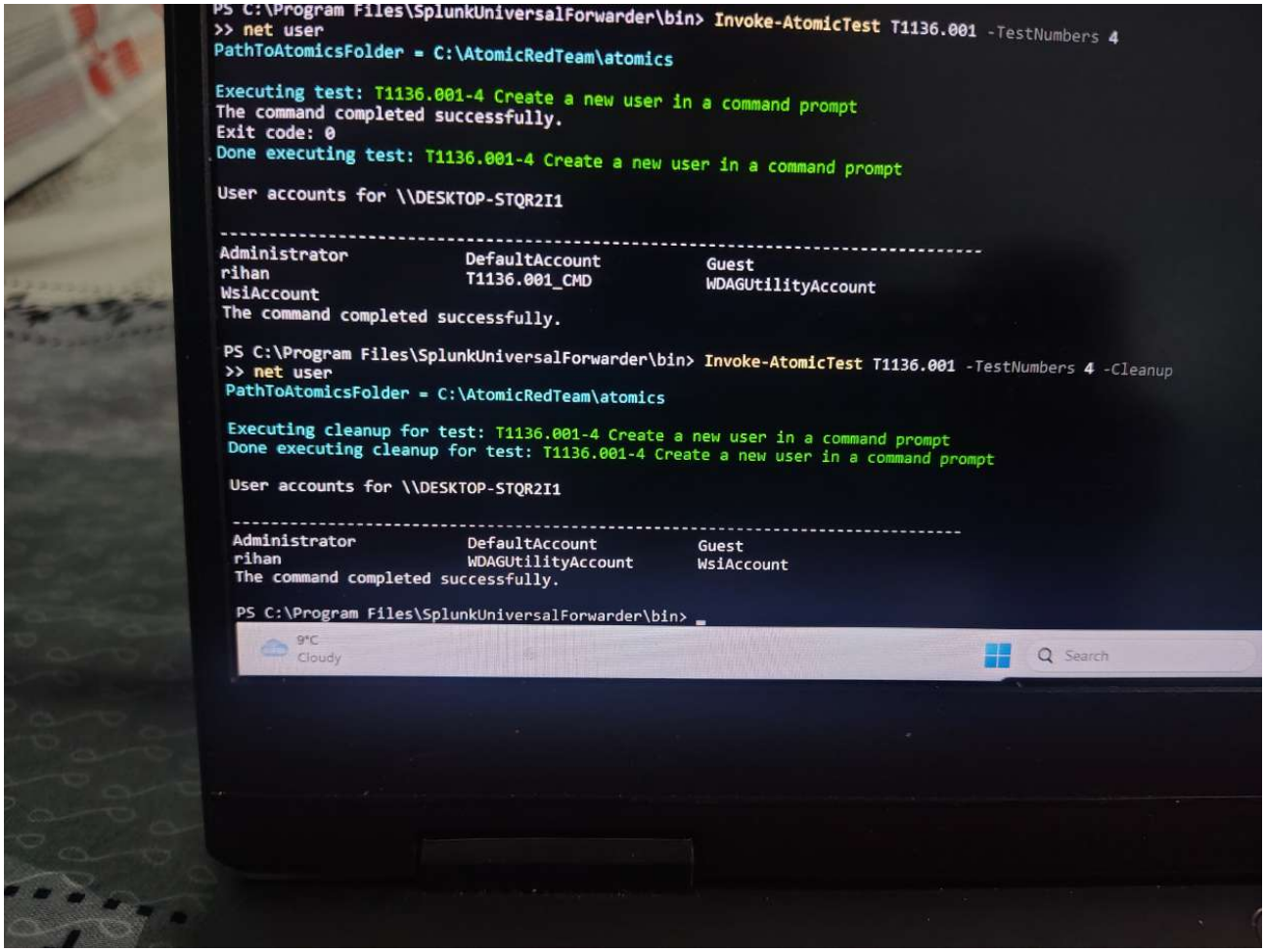


Figure 4.2: T1136.001-4 creates T1136.001_CMD (net user), then cleanup removes it.

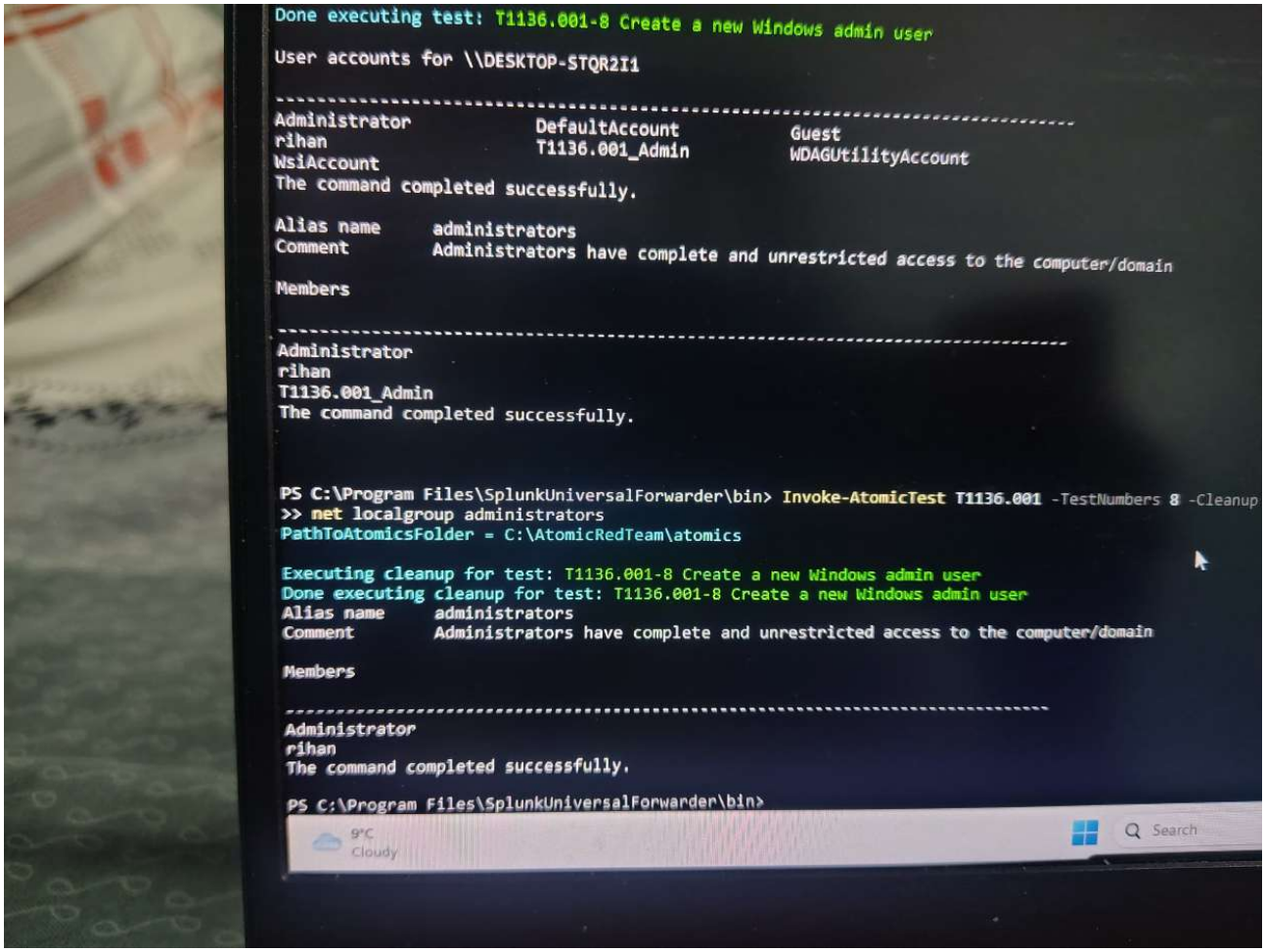


Figure 4.3: T1136.001-8 creates T1136.001_Admin, which is listed in the administrators group; cleanup then removes it.

Step 2: Detection in Splunk. A Windows Security log search for account-management events returned 9 events in time order, matching the test sequence exactly:

Time	Event ID	Meaning	Target	Actor
13:22:10	4720	User account created	T1136.001_CMD	rihan
13:22:10	4722	User account enabled	T1136.001_CMD	rihan
13:22:10	4732	Added to local group	Users	rihan
13:22:31	4726	User account deleted (cleanup)	T1136.001_CMD	rihan
13:22:55	4720 / 4722	Created / enabled	T1136.001_Admin	rihan
13:22:55	4732	Added to local group	Users, then Administrators	rihan
13:23:03	4726	User account deleted (cleanup)	T1136.001_Admin	rihan

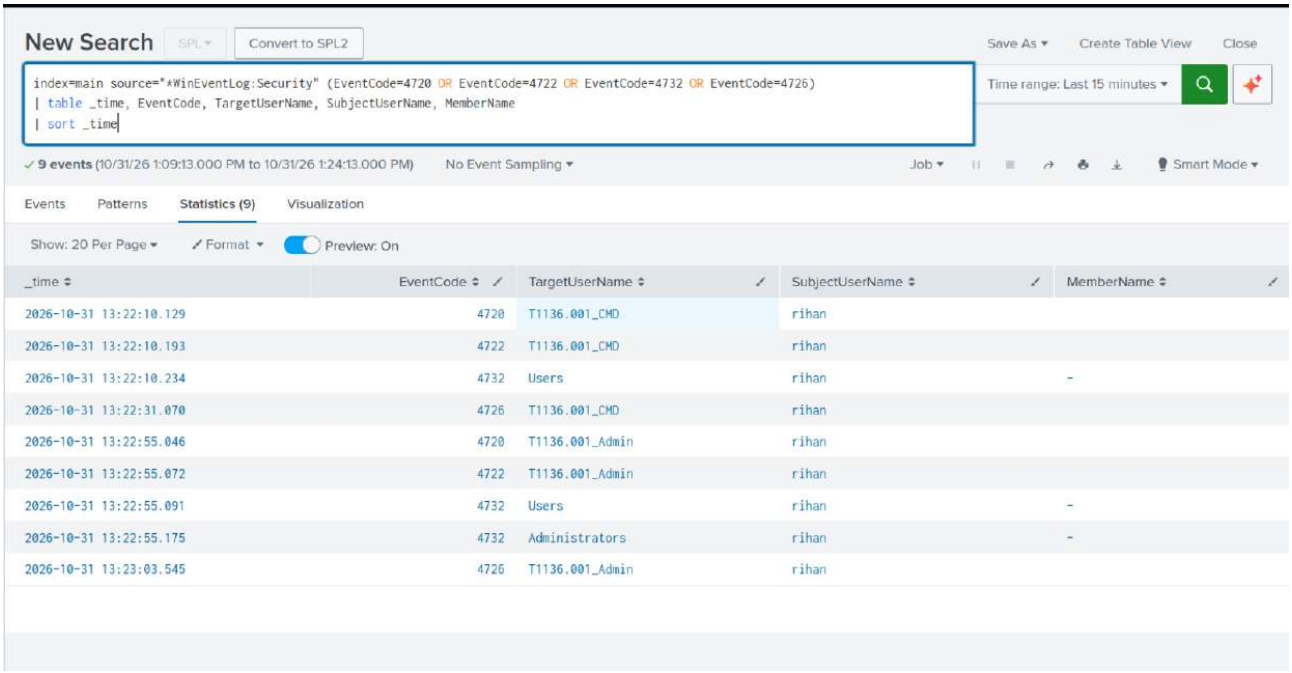


Figure 4.4: Splunk Security log search (EventCodes 4720, 4722, 4732, 4726) showing the full create, enable, group-add, delete lifecycle.

Key detection point: event 4732 adding an account to **Administrators** right after 4720 is the high-signal indicator of a privileged persistence account.